

PySpark Code

Build the base customer_spend table

```
customer_spend = (customers.join(orders, "customer_id")
    .groupBy("customer_id", "first_name", "last_name", "city")
    .agg(_sum("total_amount").alias("total_spend")))

customer_spend.createOrReplaceTempView("customer_spend")
customer_spend.orderBy(col("total_spend").desc()).show(10)
```

customer_id	first_name	last_name	city	total_spend
43	Lily	Turner	Chicago	8335.0
46	Michael	Cooper	Seattle	4000.0
36	Mason	Campbell	Baltimore	3435.0
7	Isabella	Davis	Boise	3385.0
49	Elena	Gray	Detroit	3290.0
26	Matthew	Allen	Milwaukee	3040.0
25	Harper	Hall	Honolulu	2730.0
41	Penelope	Roberts	Las Vegas	2720.0
50	Henry	Phillips	Indianapolis	2450.0
47	Amelia	James	San Antonio	2280.0

only showing top 10 rows

Methods to perform :

Method 1 — Conditional logic segmentation

```
segmented = customer_spend.withColumn(  
    "segment",  
    when(col("total_spend") > 2000, "Gold")  
    .when((col("total_spend") >= 500) & (col("total_spend") <= 2000), "Silver")  
    .otherwise("Bronze")  
).orderBy(col("total_spend").desc())  
  
segmented.show(10)
```

```
... +-----+-----+-----+-----+-----+-----+  
|customer_id|first_name|last_name|city|total_spend|segment|  
+-----+-----+-----+-----+-----+-----+  
|43|Lily|Turner|Chicago|8335.0|Gold|  
|46|Michael|Cooper|Seattle|4000.0|Gold|  
|36|Mason|Campbell|Baltimore|3435.0|Gold|  
|7|Isabella|Davis|Boise|3385.0|Gold|  
|49|Elena|Gray|Detroit|3290.0|Gold|  
|26|Matthew|Allen|Milwaukee|3040.0|Gold|  
|25|Harper|Hall|Honolulu|2730.0|Gold|  
|41|Penelope|Roberts|Las Vegas|2720.0|Gold|  
|50|Henry|Phillips|Indianapolis|2450.0|Gold|  
|47|Amelia|James|San Antonio|2280.0|Gold|  
+-----+-----+-----+-----+-----+-----+  
only showing top 10 rows
```

Bucketizer (MLlib)

```
from pyspark.ml.feature import Bucketizer  
  
splits = [-float("inf"), 500, 2000, float("inf")]  
bucketizer = Bucketizer(splits=splits, inputCol="total_spend", outputCol="bucket_index")  
bucketed = bucketizer.transform(customer_spend)  
  
bucketed = bucketed.withColumn(  
    "segment_from_bucketizer",  
    when(col("bucket_index") == 0, "Bronze")  
    .when(col("bucket_index") == 1, "Silver")  
    .otherwise("Gold")  
)  
bucketed.select("customer_id", "total_spend", "bucket_index", "segment_from_bucketizer").show(10)
```

```
+-----+-----+-----+-----+  
|customer_id|total_spend|bucket_index|segment_from_bucketizer|  
+-----+-----+-----+-----+  
|14|1635.0|1.0|Silver|  
|39|1705.0|1.0|Silver|  
|19|1180.0|1.0|Silver|  
|16|390.0|0.0|Bronze|  
|18|360.0|0.0|Bronze|  
|28|180.0|0.0|Bronze|  
|36|3435.0|2.0|Gold|  
|5|1100.0|1.0|Silver|  
|37|1005.0|1.0|Silver|  
|20|220.0|0.0|Bronze|  
+-----+-----+-----+-----+  
only showing top 10 rows
```

Window-based ranking

```
from pyspark.sql.functions import percent_rank, col, when
from pyspark.sql.window import Window

window = Window.orderBy("total_spend")
ranked = customer_spend.withColumn("spend_percentile", percent_rank().over(window))

ranked_segmented = ranked.withColumn(
    "segment_rank_based",
    when(col("spend_percentile") >= 0.66, "Gold")
    .when(col("spend_percentile") >= 0.33, "Silver")
    .otherwise("Bronze")
).orderBy(col("total_spend").desc())

ranked_segmented.show(10)
```

customer_id	first_name	last_name	city	total_spend	spend_percentile	segment_rank_based
43	Lily	Turner	Chicago	8335.0	1.0	Gold
46	Michael	Cooper	Seattle	4000.0	0.9772727272727273	Gold
36	Mason	Campbell	Baltimore	3435.0	0.9545454545454546	Gold
7	Isabella	Davis	Boise	3385.0	0.9318181818181818	Gold
49	Elena	Gray	Detroit	3290.0	0.9090909090909091	Gold
26	Matthew	Allen	Milwaukee	3040.0	0.8863636363636364	Gold
25	Harper	Hall	Honolulu	2730.0	0.8636363636363636	Gold
41	Penelope	Roberts	Las Vegas	2720.0	0.8409090909090909	Gold
50	Henry	Phillips	Indianapolis	2450.0	0.8181818181818182	Gold
47	Amelia	James	San Antonio	2280.0	0.7954545454545454	Gold

only showing top 10 rows

Practice Tasks

Task 1 & 2 -Count customers per segment

```
segmented.groupBy("segment").agg(count("*").alias("customer_count")).show()
```

segment	customer_count
Silver	18
Gold	12
Bronze	15

Task 3 - Quantile-based segmentation

```
▶ quantiles = customer_spend.approxQuantile("total_spend", [0.33, 0.66], 0.0)
low, high = quantiles
print("cutoffs:", low, high)

quantile_segmented = customer_spend.withColumn(
    "segment_quantile",
    when(col("total_spend") > high, "Gold")
    .when((col("total_spend") > low) & (col("total_spend") <= high), "Silver")
    .otherwise("Bronze")
)
quantile_segmented.groupBy("segment_quantile").agg(count("*").alias("customer_count")).show()
```

```
*** cutoffs: 445.0 1635.0
+-----+-----+
|segment_quantile|customer_count|
+-----+-----+
|      Silver    |      15      |
|      Gold      |      15      |
|      Bronze    |      15      |
+-----+-----+
```

Task 4 - Compare all methods

```

comparison = (segmented.select("customer_id", "total_spend", "segment")
              .join(quantile_segmented.select("customer_id", "segment_quantile"), "customer_id")
              .join(ranked_segmented.select("customer_id", "segment_rank_based"), "customer_id"))

comparison.orderBy(col("total_spend").desc()).show(50, truncate=False)

```

```

*** +-----+-----+-----+-----+-----+
|customer_id|total_spend|segment|segment_quantile|segment_rank_based|
+-----+-----+-----+-----+-----+
|43          |8335.0     |Gold   |Gold            |Gold              |
|46          |4000.0     |Gold   |Gold            |Gold              |
|36          |3435.0     |Gold   |Gold            |Gold              |
|7           |3385.0     |Gold   |Gold            |Gold              |
|49          |3290.0     |Gold   |Gold            |Gold              |
|26          |3040.0     |Gold   |Gold            |Gold              |
|25          |2730.0     |Gold   |Gold            |Gold              |
|41          |2720.0     |Gold   |Gold            |Gold              |
|50          |2450.0     |Gold   |Gold            |Gold              |
|47          |2280.0     |Gold   |Gold            |Gold              |
|44          |2200.0     |Gold   |Gold            |Gold              |
|38          |2100.0     |Gold   |Gold            |Gold              |
|45          |1950.0     |Silver |Gold            |Gold              |
|40          |1820.0     |Silver |Gold            |Gold              |
|39          |1705.0     |Silver |Gold            |Gold              |
|14          |1635.0     |Silver |Silver          |Silver            |
|1           |1600.0     |Silver |Silver          |Silver            |
|31          |1450.0     |Silver |Silver          |Silver            |
|24          |1445.0     |Silver |Silver          |Silver            |
|17          |1300.0     |Silver |Silver          |Silver            |
|19          |1180.0     |Silver |Silver          |Silver            |
|12          |1180.0     |Silver |Silver          |Silver            |
|11          |1150.0     |Silver |Silver          |Silver            |
|5           |1100.0     |Silver |Silver          |Silver            |
|37          |1005.0     |Silver |Silver          |Silver            |
|48          |800.0      |Silver |Silver          |Silver            |
|3           |675.0      |Silver |Silver          |Silver            |
|13          |580.0      |Silver |Silver          |Silver            |
|6           |580.0      |Silver |Silver          |Silver            |
|23          |500.0      |Silver |Silver          |Silver            |
|8           |445.0      |Bronze |Bronze          |Bronze            |
|16          |390.0      |Bronze |Bronze          |Bronze            |
|18          |360.0      |Bronze |Bronze          |Bronze            |
|2           |300.0      |Bronze |Bronze          |Bronze            |
|9           |290.0      |Bronze |Bronze          |Bronze            |
|20          |220.0      |Bronze |Bronze          |Bronze            |
|33          |220.0      |Bronze |Bronze          |Bronze            |
|28          |180.0      |Bronze |Bronze          |Bronze            |
|10          |180.0      |Bronze |Bronze          |Bronze            |
|22          |160.0      |Bronze |Bronze          |Bronze            |
|42          |135.0      |Bronze |Bronze          |Bronze            |
|15          |125.0      |Bronze |Bronze          |Bronze            |
|29          |105.0      |Bronze |Bronze          |Bronze            |
|21          |80.0       |Bronze |Bronze          |Bronze            |
|30          |25.0       |Bronze |Bronze          |Bronze            |
+-----+-----+-----+-----+-----+

```

